

FORGING IMAGE WATERMARKS BY SCHEME IDENTIFICATION AND NATIVE RE-ENCODING

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Team V • Code: <https://github.com/SaadH-077/TML-Watermark-Forging>

1 INTRODUCTION

We are given 8 unknown image watermarking schemes $WM_1, \dots, WM_8$. For each we receive 25 watermarked “carrier” images that all carry *one* shared hidden message, and we must transplant that watermark onto 25 assigned clean target images so a hidden detector recovers the message while each forged image stays perceptually close to its target. Carriers and targets are different images, so this is true transplantation rather than re-stamping the same picture, and the resolution is fixed per scheme. A submission is a flat archive of 200 PNGs, one per target.

The score is $S_{\text{final}} = S_{\text{det}} \cdot S_{\text{qit}}$, with $S_{\text{det}} = \max(0, 2(\text{bitacc} - 0.5))$ and $S_{\text{qit}} = \exp(-8 \text{LPIPS})$. Two properties decide the design. First, S_{det} is a hard gate: bit-accuracy 0.5 zeroes the image no matter how good it looks. Second, LPIPS is measurable locally, but bit-accuracy is not, because we hold no detector; the only detection signal is the leaderboard (one submission per hour, best kept, 30% public). So detection strength, not visual quality, is the binding constraint, and our rule was to land the bits first and recover LPIPS second. Our final submission reaches a public score of **0.8553**.

2 APPROACH

Identification beats blind transplant. The averaging attack estimates the watermark as the mean carrier residual and adds it to the target [1], but fails on frequency-domain and content-adaptive marks. On the first two schemes where we could also decode, a native re-encode reached $S_{\text{final}} = 0.89$ (WM_1) and 0.86 (WM_2) while an additive transplant scored ≈ 0 . The better route is to *identify* the scheme, decode the shared message, and re-embed it with the scheme’s own genuine encoder: the forgery is then a real watermark, decoding at bit-accuracy 1.0 and generalizing across the split. Identification became our main strategy.

The oracle test. Because the 25 carriers share one message, we can identify a scheme without a detector: decode all 25 with a candidate decoder, and if they agree on one message (carrier-agreement near 1.0) while clean images decode randomly (near 0.5), it is identified. We accepted a match only with a *validated positive control* (a genuine encode-then-decode round trip proving the decoder reads real marks) plus a known-negative control from an already identified scheme. This mattered, because several apparent matches were artifacts, a decoder returning constant output regardless of input, or content correlation that also lifts clean images. Running classical decoders on WM_3 ’s chroma channels, for instance, reports carrier-agreement 1.000, but the same decoder returns identical constant bits on pure random noise, a degenerate reading, not a recovered message.

What we did for each watermark. We identified all eight schemes and re-encoded each natively at round-trip bit-accuracy 1.0 (Table 1): dwtDct (WM_1) and RivaGAN (WM_2) from invisible-watermark [2], VINE-R (WM_4) [7], CIN (WM_5) [4], MBRS (WM_6) [3], and two TrustMark variants [6] (WM_7 , WM_8). Two details mattered: WM_1 is a quantization mark, so we amplify its residual at $s = 1.5$ to widen the margin (see the robustness audit below), and WM_8 matched only under TrustMark variant P with error correction disabled, which is why an earlier sweep of variant Q missed it. WM_3 , the hardest, is ArtificialGANFingerprints [8] and is discussed separately below.

Table 1: Per-scheme forge, all genuine native re-encodes (round-trip bit-accuracy 1.0; WM₃ 0.96). LPIPS / S_{final} are local estimates; WM₃ spans the AlexNet/VGG backbones.

WM	Scheme (message)	Forge	LPIPS / S_{final}
WM ₁	dwtDct, 30-bit	native re-encode, residual amplified $s=1.5$	0.006 / 0.89
WM ₂	RivaGAN, 32-bit	native re-encode	0.016 / 0.86
WM ₃	ArtificialGANFingerprints, 100-bit	native re-embed (256^2)	0.024 / 0.67–0.78
WM ₄	VINE-R, 100-bit	native re-encode (256^2)	0.003 / 0.94
WM ₅	CIN, 30-bit	native re-encode (128^2)	0.001 / 0.99
WM ₆	MBRS, 256-bit	native re-encode (256^2)	0.002 / 0.96
WM ₇	TrustMark-Q, 61-bit	native re-encode (512^2)	0.001 / 0.99
WM ₈	TrustMark-P (no ECC), 100-bit	native re-encode (512^2)	0.001 / 0.99

WM₃: identified as ArtificialGANFingerprints. WM₃ resisted the averaging attack because it is content-adaptive: the shared residual sits at the clean-image noise floor (cross-carrier correlation ≈ 0.001), so there is no fixed pattern to average out, and a blind transplant caps near $S_{\text{final}} \approx 0.43$. We ruled out about 27 public decoders and mistook the remaining gap for a structural ceiling. The decisive step was a decode-consistency sweep at the native 256^2 : since all 25 carriers share one message, the *correct* decoder returns the same bits on all of them. Most candidates gave chance, but the ArtificialGANFingerprints [8] AFHQ cat2dog 256^2 checkpoint, a StegaStamp [9] autoencoder, read the 25 carriers at agreement 0.9996 while clean images sat at 0.48, which is only possible if the benchmark used these exact weights, so the hidden detector *is* this decoder. (Earlier StegaStamp tests used only its 400^2 checkpoint and were excluded on resolution.) We majority-vote the 100-bit message and re-embed it with the matching encoder, forged = clip($y + ar$), $r = \text{Enc}(\text{msg}, y) - y$, decoding at mean bit-accuracy 0.96. LPIPS is the only cost and depends on the grader’s backbone; since that is unknown, we set each a to maximise the *worst case* over AlexNet and VGG, not the AlexNet optimum, which hides high-frequency texture that VGG penalises.

Native robustness locates the residual gap. The leaderboard grades bit-accuracy after an unknown pre-detection transform, so clean-PNG bit-accuracy overestimates the server. A local robustness oracle (decode, apply JPEG/resize/noise/blur, decode again) shows six of the seven learned natives already robust; only WM₁, the classical dwtDct mark, is fragile (0.60, collapsing under JPEG). Strengthening its margin was a wash (+0.0002): the quality cost cancelled the detection gain, since dwtDct is JPEG-fragile at every strength. The natives sit near their ceiling under these public encoders, an informative negative result rather than a missed lever.

3 RESULTS

Adding schemes one change at a time, the public score climbed 0.6613 (3 native) $\rightarrow$ 0.7188 (+CIN, MBRS) $\rightarrow$ 0.8211 (+VINE-R, TrustMark-P, 7 native + WM₃ transplant) $\rightarrow$ 0.8274 (chroma WM₃ + masked natives) $\rightarrow$ 0.8364 (WM₃ color-axis transplant + WM₁ amplification $s=1.5$) $\rightarrow$ **0.8553** (WM₃ re-embedded natively as ArtificialGANFingerprints). Every gain came from a per-scheme method that generalizes across the split, not from tuning to the public sample; the final +0.019 changed only the WM₃ set, so it is entirely attributable to the identification. With all eight schemes native, the remaining gap to the strongest teams is the quality and robustness of the public encoders, not a missing scheme. The best submission rebuilds with one command (`build_best.py`), and every reported number has a script behind it.

4 CONCLUSION

Watermark forgery under near black-box conditions is practical and cheap: for any deployed public scheme, an attacker with a handful of same-message watermarked images can identify the scheme and re-encode with its own public encoder, a genuine, generalizing forgery that stamps arbitrary content as authentically watermarked. WM₃ is the sharpest lesson. As a content-adaptive learned mark it defeats the averaging forgery that breaks the additive schemes, and we briefly mistook that resistance for security. But content-adaptivity is not protection once the weights are public: the per-image embedding that hides the mark is exactly what its own encoder reproduces on any target. Nor is obscurity, WM₁ and WM₅ fall to plain averaging and WM₃ fell the moment we matched it to a public checkpoint. Reliable provenance has to rest on a secret key or detector, not on imperceptibility, content-adaptivity, or an undisclosed scheme built on public encoders.

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